1. The trait for medium-sized leaves in iris is determined by the genetic condition PP'. Plants with large leaves are PP, whereas plants with small leaves are P'P'. The trait for red flowers is controlled by the genes RR, pink by RR', and white by R'R'. A cross is made between two plants each with medium-sized leaves and pink flowers. If they produce 320 seedlings, what would be the expected phenotypes, and in what numbers would they be expected? Assume no linkage.

$PP'RR' \times PP'RR'$

Take $RR' \times RR'$ as an example:

	R	R'
R	RR	RR'
R'	RR'	R'R'

That is with $\frac{1}{4}$ red, $\frac{1}{2}$ pink and $\frac{1}{4}$ white. Thus:

large leaves and red flowers: $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. $320 \times \frac{1}{16} = 20$

medium, red: $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$. $320 \times \frac{1}{8} = 40$

small, red: $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. $320 \times \frac{1}{16} = 20$

large, pink: $\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$. $320 \times \frac{1}{8} = 40$.

medium, pink: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. $320 \times \frac{1}{4} = 80$

small, pink: $\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$. $320 \times \frac{1}{8} = 40$.

large, white: $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. $320 \times \frac{1}{16} = 20$.

medium, white: $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$. $320 \times \frac{1}{8} = 40$.

small, white: $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. $320 \times \frac{1}{16} = 20$.

2. In a species of tropical fish, a colorful orange and black variety called montezuma occurs. When two montezumas, are crossed, 2/3 of the progeny are montezuma, and 1/3 are the wild-type, dark grayish green color. Montezuma is a single-gene trait, and montezuma fish are never true-breeding.

a. Explain the inheritance pattern seen here and show how your explanation accounts for the phenotypic ratios given.

b. In this same species, the morphology of the dorsal fin is altered from normal to ruffled by homozygosity for a recessive allele designated *f*. What progeny would you expect to obtain, and in what proportions, from the cross of a Montezuma fish homozygous for normal fins to a green, ruffled fish?

c. What phenotypic ratios of progeny would be expected from the crossing of two of the Montezuma progeny from part *b* ?

a. Montezuma is indicated by M, and WT is indicated by m.
The ratio means that the parents were Mm, and MM or mm is lethal.

	M	m
M	MM	Mm
m	Mm	mm

$\Rightarrow \frac{2}{3}$ Montezuma, $\frac{1}{3}$ WT.

b. ① if mm is lethal, then: MmFF × MMff.
proportions:
MMFf: $\frac{1}{2} \times 1 = \frac{1}{2}$. greenish, normal fin
MmFf: $\frac{1}{2} \times 1 = \frac{1}{2}$. Montezuma, normal fin

② if MM is lethal, then: MmFF × mmff.
proportions:
MmFf: $\frac{1}{2} \times 1 = \frac{1}{2}$. greenish, normal fin
mmFf: $\frac{1}{2} \times 1 = \frac{1}{2}$. Montezuma, normal fin

c. ① MMFf × MmFf (mm is lethal)

greenish, normal fin (MMF_) : $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

Montezuma, normal fin (MmF_) : $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

greenish, ruffled fin (MMff) : $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

Montezuma, ruffled fin (Mmff) : $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

② MmFf × mmFf (MM is lethal)

greenish, normal fin (mmF_) : $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

Montezuma, normal fin (MmF_) : $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

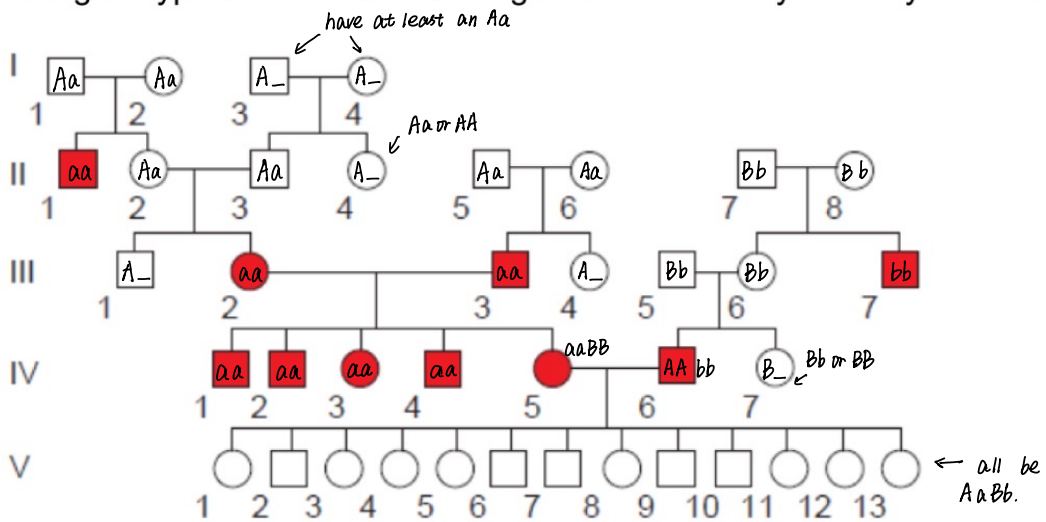
greenish, ruffled fin (mmff) : $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

Montezuma, ruffled fin (Mmff) : $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

3. Filled-in symbols in the pedigree that follows designate individuals suffering from deafness.

a. Study the pedigree and explain how deafness is being inherited.

b. What is the genotype of the individuals in generation V? Why are they not affected?



a. autosomal recessive inheritance.

Explain: From I_1、I_2 → II_1、II_2 & II_5、II_6 → III_3、III_4 & II_7、II_8 → III_6、III_7,

It's easy to find out that it's recessive inheritance.

Because IV_{1-5} all have diseases, so both III_2 & III_3 are aa.

The genotype of each person is illustrated above.

If it's with X chromosome, I_1: X^AY, I_2: X^AX^a, so II_1: X^aY, II_2: X^AX^a, III_2: X^aX^a, so II_3 should be X^aY, but II_3 is healthy, so it's contradict.

If it's with Y chromosome, then III_1: XY^a, I_1 should be XY^a, contradict.

From above, it's autosomal recessive inheritance.

b. Because V_{1-13} are all healthy, so IV_6 family have another virulence gene on another chromosome. (indicated by b). So IV_5 is aaBB, IV_6 is AAbb, their progency are AaBb, all are healthy.